

■■ Kubernetes Learning Plan via Mini Projects (12 Weeks)

Phase 1 – Foundations (Weeks 1–3)

Goal: Get familiar with core Kubernetes objects by deploying simple apps.

Project 1 – Hello Kubernetes (Week 1)

- Deploy Nginx Pod, expose via Service (NodePort), scale with Deployment

Skills: Pods, Deployments, Services, scaling

Project 2 – Configured Web App (Week 2)

- Deploy backend app (Node.js or Spring Boot)

- Use ConfigMap for configs, Secret for DB password

Skills: ConfigMaps, Secrets, app configuration

Project 3 – Persistent Database (Week 3)

- Deploy Postgres database with PersistentVolume + PVC

- Connect backend app to Postgres

Skills: Persistent storage, app-to-DB communication

Phase 2 – Intermediate (Weeks 4–6)

Goal: Build a complete multi-tier app with networking and background tasks.

Project 4 – Full-Stack Application (Week 4)

- Deploy frontend + backend + database

- Expose frontend via Ingress controller

- Use ClusterIP services internally

Skills: Multi-tier apps, Ingress, service discovery

Project 5 – Health & Security (Week 5)

- Add readiness/liveness probes to backend

- Restrict DB access with NetworkPolicy

Skills: Pod health checks, network policies

Project 6 – Scheduled Jobs (Week 6)

- Add CronJob to clean DB daily

- Add resource requests/limits for Pods

Skills: Jobs, CronJobs, resource management

Phase 3 – Advanced (Weeks 7–9)

Goal: Add monitoring, autoscaling, and security best practices.

Project 7 – Logging & Monitoring with EFK (Week 7)

- Deploy Fluentbit as DaemonSet to collect logs

- Send logs to Elasticsearch

- Deploy Kibana to visualize logs

- Install Prometheus & Grafana for metrics

Deliverable: Centralized logging (EFK) + monitoring dashboards

Skills: EFK stack, observability

Project 8 – Autoscaling (Week 8)

- Deploy load testing tool (siege/k6)
 - Configure HPA for backend
 - Add PodDisruptionBudget for resilience
- Skills: Autoscaling apps, resiliency

Project 9 – RBAC & Secrets Management (Week 9)

- Create read-only developer role with RBAC
 - Secure secrets with SealedSecrets or Vault
- Skills: RBAC, secrets management

Phase 4 – Mastery (Weeks 10–12)

Goal: Production-level deployments with GitOps and service mesh.

Project 10 – GitOps Pipeline (Week 10)

- Install ArgoCD
 - Store manifests in GitHub, auto-deploy via GitOps
- Skills: GitOps, CI/CD

Project 11 – Service Mesh (Week 11)

- Install Istio or Linkerd
 - Add canary deployment (10% traffic to backend v2)
- Skills: Service mesh, traffic shifting

Project 12 – Cloud Cluster (Week 12)

- Deploy cluster on AWS EKS (or GKE/AKS)
 - Enable Cluster Autoscaler
 - Backup/restore DB and etcd
- Skills: Cloud Kubernetes, node autoscaling, disaster recovery

Final Capstone Project (Week 12+)

- Multi-tier microservices app
- Ingress + Service Mesh
- CI/CD with ArgoCD
- Monitoring (Prometheus, Grafana)
- Logs (EFK: Fluentbit, Elasticsearch, Kibana)
- Tracing (Jaeger)
- Security (RBAC, NetworkPolicies, Secrets)
- Autoscaling (HPA + Cluster Autoscaler)
- Cloud deployment (EKS/GKE/AKS)